

Probability, Statistics & Data Analysis

3: Discrete Random Variables

**The generation of random numbers
is too important to be left to chance**

Robert R. Coveyou

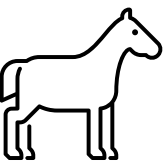
3: Discrete Random Variables

What we will cover

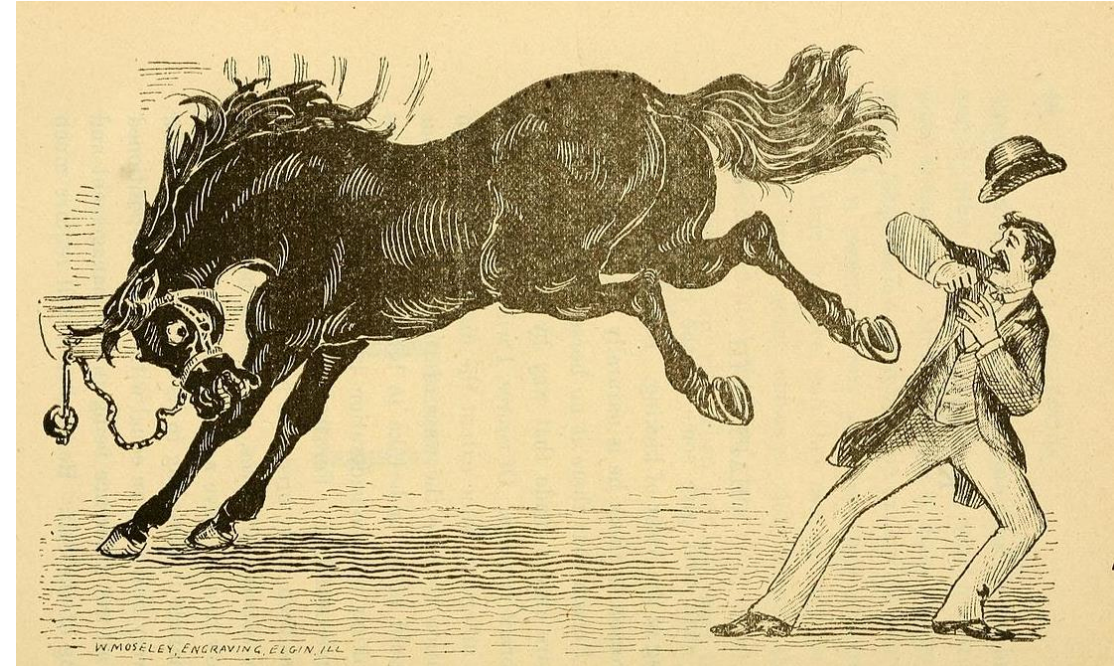
- Introduction to RVs
- Probability Mass Function
- Cumulative Distribution Functions
- RV Descriptive Statistics
 - Expected Value
 - Variance
- Binomial Distribution
- Poisson Distribution



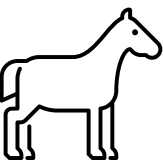
Vengeful Horses (?)



- Prussian Army in 1889 had a problem with soldiers being fatally kicked by horses
- The data were from 10 army corps over 20 years
 - Due to the unfortunate similarity between ‘corps’ and ‘corpse’, we will use the term Division (a corps is technically 2-5 divisions, but let’s not dwell on that)
- We therefore have 200 Division-Years (200 data points)



Vengeful Horses – Data

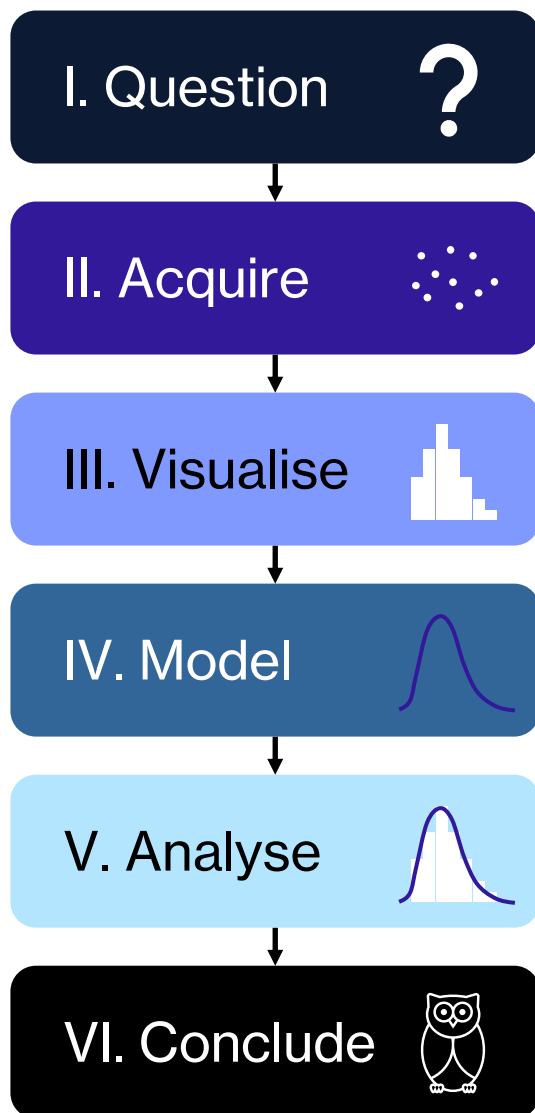


- N is the number of Division-Years
- X is number of deaths by horse
- Hence
 - 3 soldiers died in a given division three times in the 20 year period
 - We do not know if this was the same division or not
 - 4 soldiers died in a given division only once in the 20 year period

Question: were the divisions in which 3 or 4 soldiers died at fault, or could this expected to occur sometimes?

X	N
0	109
1	65
2	22
3	3
4	1

Framework – Vengeful Horses

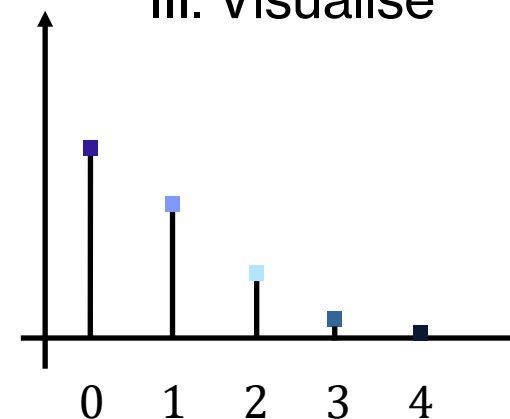


I. Is someone at fault for the deaths-by-horse?

II. Data

X	N
0	109
1	65
2	22
3	3
4	1

III. Visualise



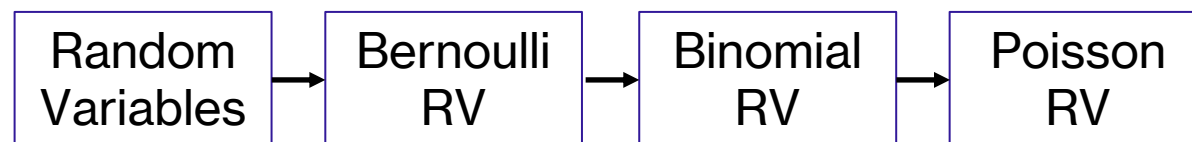
IV. The data are sampled from a Poisson distribution

– Check if the data are consistent with this model

V. If they are, estimate how frequently we might expect 3 or 4 deaths-by-horse in a given division-year

VI. Compare these numbers to our data and draw a conclusion

But first.....



X 

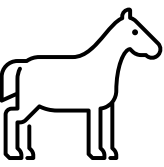
Random Variables

Intro to Random Variables (RV)



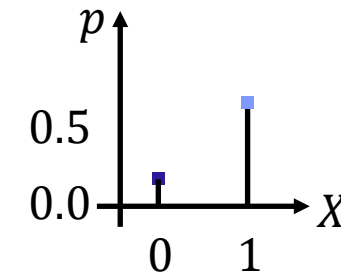
- RV: a rule that associates a number with an outcome in the sample space, $\mathcal{S}$
- Bernoulli (binary) RV
 - Only has values of 0 or 1
 - E.g. Dead or Alive
- Discrete RV
 - Only take finite numbers
 - E.g. Number of WBCs in a blood sample
 - Note binning (as in histogram) can yield discrete RV from continuous input
- Continuous RV
 - Can exist anywhere on a continuous number line
 - E.g. Length of a certain shark species

Types of Random Variables



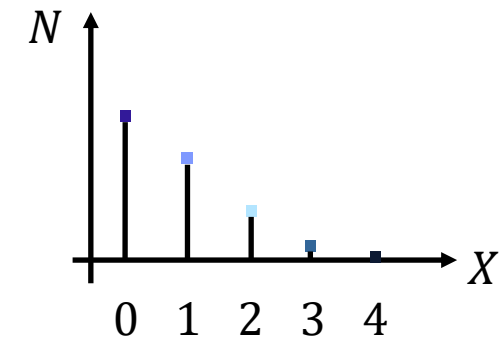
Bernoulli: is a soldier killed by a kick?

- Outcomes
 - A: Soldier is killed by a horse kick
 - B: Soldier is not killed by a horse kick
- As RV
 - $X(A) = 1$
 - $X(B) = 0$



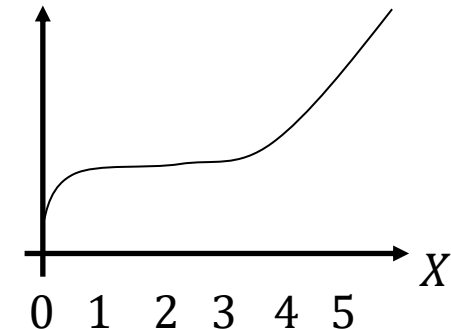
Discrete: number of deaths per year for a given Division

- As RV
 - X = number of deaths in a year for a given division
 - Discrete Random Variable



Continuous: how long does a soldier survive after a kick?

- As RV
 - X = time in hours of survival after being kicked
 - Continuous Random Variable





Probability Mass Functions

Probability distributions for discrete variables

Probability Distributions for Discrete RVs

- If we prescribe probabilities for an RV, X , then we can produce **probability distributions**
- Known as **Probability Mass Function, PMF**, for Discrete RV
- Defines how the total probability of 1 is shared between possible values of X

$$p(x) = P(X = x)$$

X is the RV

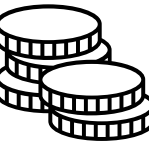
x is a specific realisation of X

$p(x)$ is the PMF

- Strictly
$$P(X = x) = P(\text{all } y \in \mathcal{S} \text{ for which } X(y) = x)$$

$\in$: is an element of
- Probability of all of the events y in the sample space $\mathcal{S}$ for which the random variable X is x
 - Because multiple events can result in the same x
 - E.g. Double Coin Flip with A=at least one tail

PMF for Coin Flip



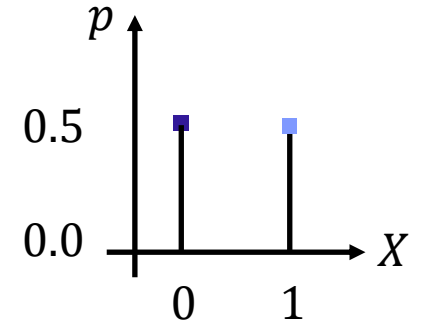
Single Coin Flip {H,T}

- Let X equal the number of tails

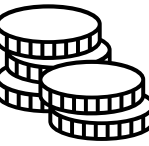
$$\text{Heads: } p(0) = P(X = 0) = P(H) = \frac{1}{2}$$

$$\text{Tails: } p(1) = P(X = 1) = P(T) = \frac{1}{2}$$

- Note: we could have defined X as the number of heads if we wanted to
- It is our choice how to define the RV
- Only two outcomes
- This is a Bernoulli RV



General form of Bernoulli RV PMF



$$p(x; \alpha) = \begin{cases} 1 - \alpha & \text{if } x = 0 \\ \alpha & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

Single Coin Flip {H,T}

$$\alpha = 0.5$$

α is known as a '**parameter**' of the distribution – each value of it can generate a different PMF

Dice Throw {1,2,3,4,5,6}

E.g. Let $X = 0$ be the outcome that y is greater than 4

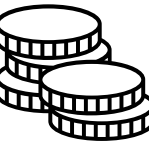
$$P(X = 0) = P(5,6) = \frac{1}{3}$$

E.g. Let $X = 1$ be the outcome that y is less than or equal to 4

$$P(X = 1) = P(1,2,3,4) = \frac{2}{3}$$

$$\alpha = \frac{2}{3}$$

PMFs for Double Coin Flip

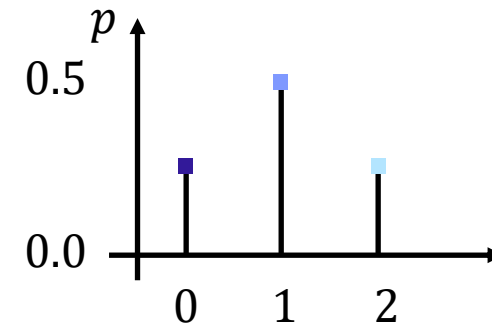


Double Coin Flip {HH, HT, TH, TT}

- Let X = the number of tails

$$\left. \begin{array}{l} X(HH) = 0 \\ X(HT) = 1 \\ X(TH) = 1 \\ X(TT) = 2 \end{array} \right\} \begin{array}{l} P(X = 0) = 0.25 \\ P(X = 1) = 0.50 \\ P(X = 2) = 0.25 \end{array}$$

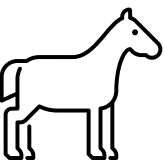
- X can take values other than 0 or 1, so it is not a Bernoulli RV



$$P(X = x) = P(\text{all } y \in \mathcal{S} \text{ for which } X(y) = x)$$

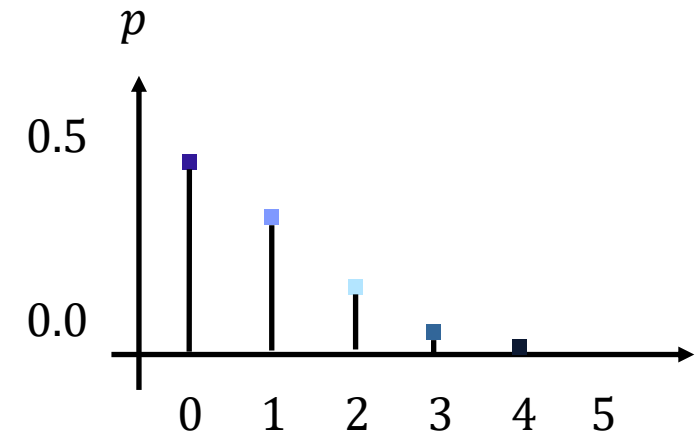
HT and TH are two events, y for which $X = 1$

PMF for Horse Kicks



Find the PMF

X	N
0	109
1	65
2	22
3	3
4	1



- If we want, e.g. $P(X < 3)$, we have to add up $P(X = 0) + P(X = 1) + P(X = 2)$
- Fine for 5 values of X , but what if we had an RV with 100 possible values of X ?



Cumulative Distribution Function

For discrete random variables

Cumulative Distribution Functions for Discrete RV

- The cumulative distribution function $F(x)$ of a discrete random variable X with probability distribution $p(x)$ is defined as

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$$

Probability that
the RV X is less
than or equal to a
specific value x

The sum of all the probabilities of
events for which y is less than x

y is dummy variable.
equivalent notation:

$$F(a) = P(X \leq a)$$

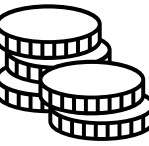
$$= \sum_{x: x \leq a} p(x)$$

Probability that
the RV X is less
than or equal to a
specific value a

CDF useful for asking about ranges for X

PMF useful for asking about specific values of X

Coin Toss CDF



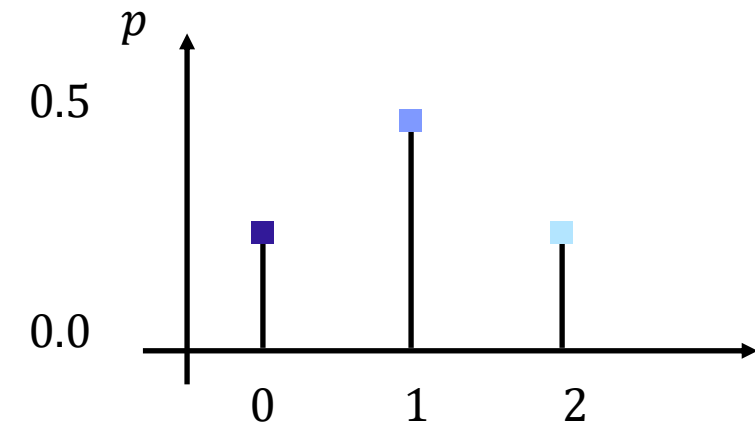
Double Coin Flip {HH, HT, TH, TT}

$$P(X = 0) = 0.25$$

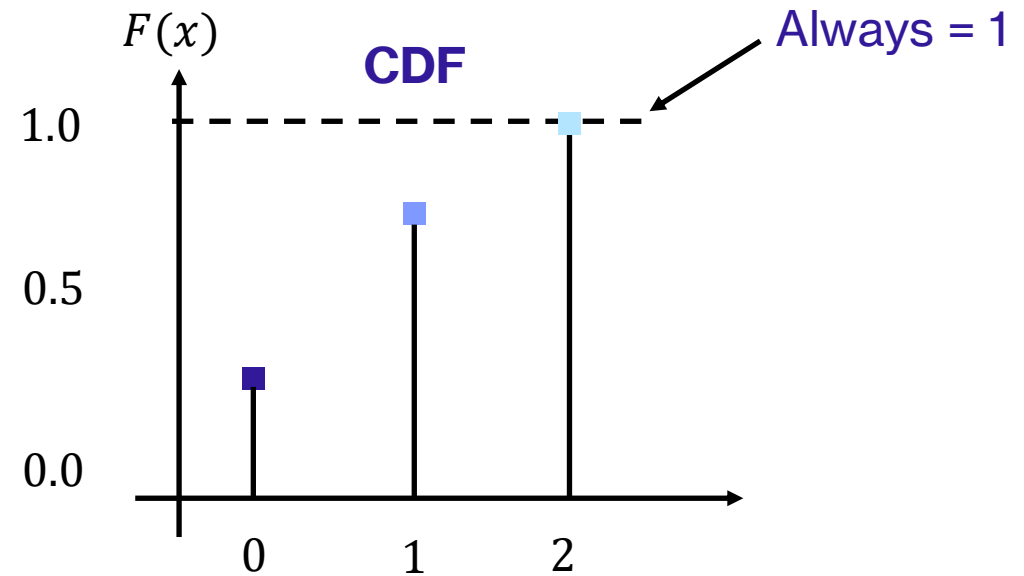
$$P(X = 1) = 0.50$$

$$P(X = 2) = 0.25$$

PMF



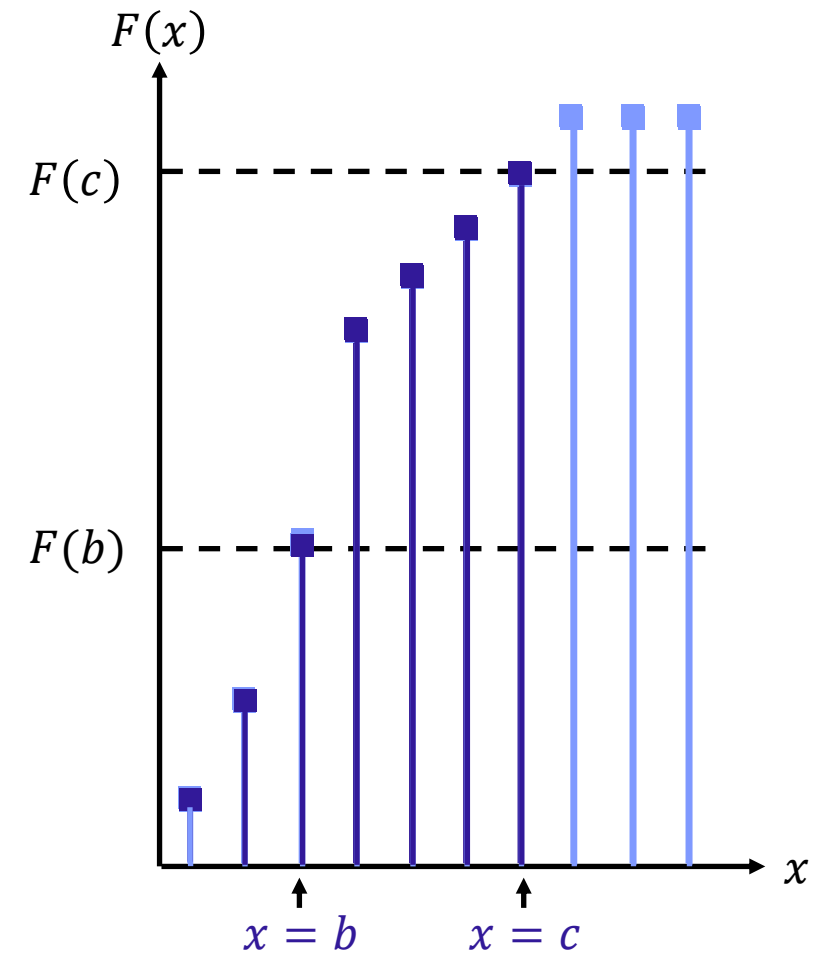
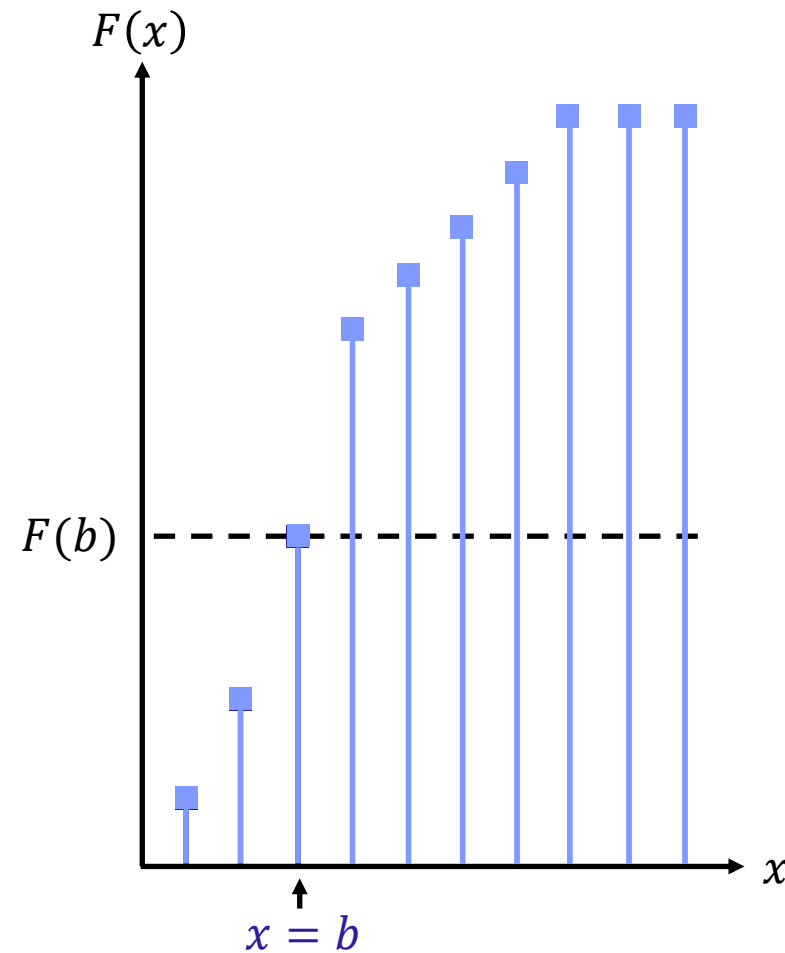
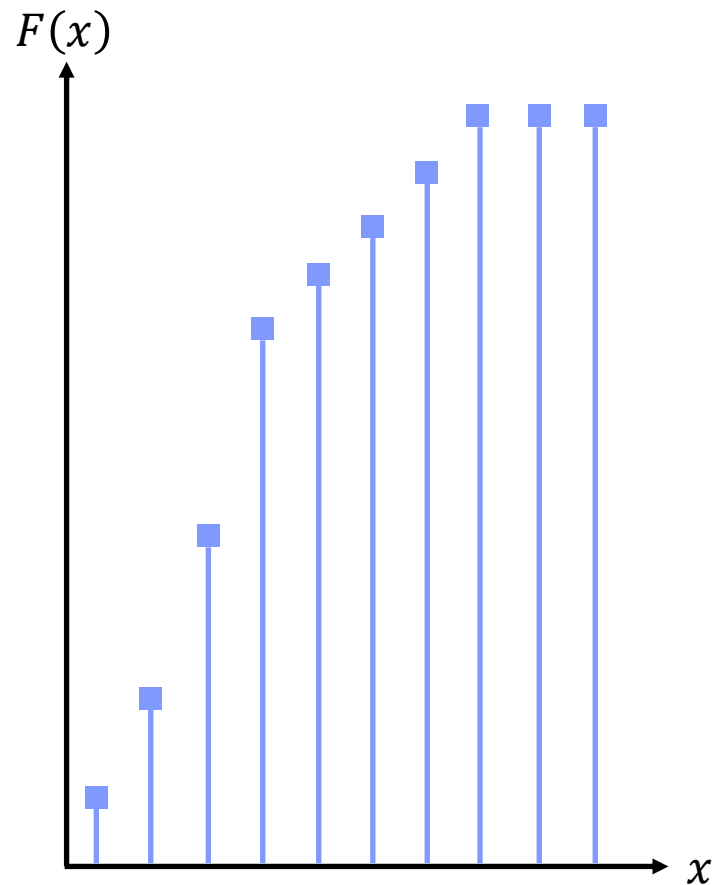
CDF



Using CDF

$$P(X \leq b) = F(b)$$

$$P(b < X \leq c) = F(c) - F(b)$$





Expected Value

For discrete random variables

Expected Value of a Discrete RV

- If X is a discrete RV with a set D of possible values and a probability mass function $p(x)$
- The expected value of X , denoted by $E(X)$ or μ_X , is

$x \in D$: for all
values of x in
the set D

$$E(X) = \mu_X = \sum_{x \in D} xp(x)$$

- Expected value is a measure of location
 - Each value gets weighted by the probability it will happen
 - Less probable events have less influence
 - If we know all $p(x)$ then it is the true population mean for our RV, hence μ_X

Expected Value of Coin Toss



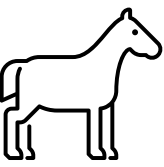
Double Coin Flip {HH, HT, TH, TT}

Let X = the number of tails

$$E(X) = \mu_X = \sum_{x \in D} xp(x)$$

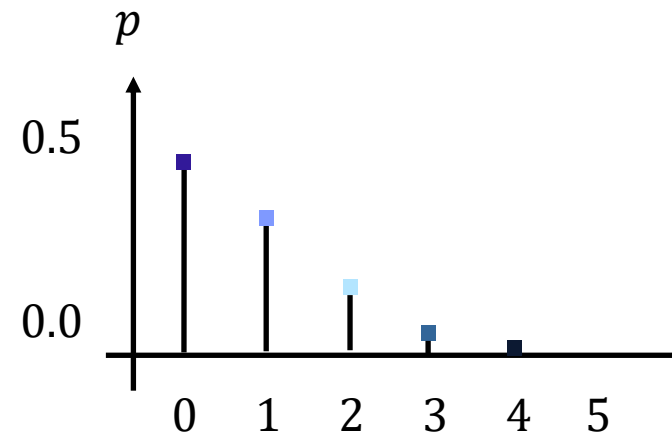
$$\left. \begin{array}{l} X(HH) = 0 \\ X(HT) = 1 \\ X(TH) = 1 \\ X(TT) = 2 \end{array} \right\} \begin{array}{l} P(X = 0) = 0.25 \\ P(X = 1) = 0.50 \\ P(X = 2) = 0.25 \end{array}$$

Expected Value of Horse Kicks



Total number of deaths by horse kick per year for a given Division

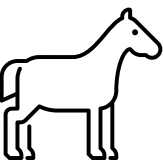
$$E(X) = \mu_X = \sum_{x \in D} xp(x)$$



X	N	p
0	109	0.545
1	65	0.325
2	22	0.110
3	3	0.015
4	1	0.005

See slide 14: these are the values from the PMF

Expected Value of Bernoulli RV



- Expected value of a Bernoulli Random Variable

$$\mu_X = \sum_{x \in D} xp(x)$$

$$\mu_X = 0 \times q + 1 \times p$$

For Bernoulli RV
 $E(X) = p$

Bernoulli: is a soldier killed by a kick?

Outcomes

A: Soldier is killed by a horse kick

B: Soldier is not killed by a horse kick

Bernoulli RV

$$X(A) = 1$$

$$X(B) = 0$$

Probabilities

$$P(A) = p$$

$$P(B) = 1 - p = q$$

X	P
0	q
1	p



Variance and Standard Deviation

For discrete random variables

Variance and SD of a Discrete RV

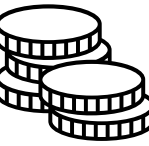
$$E(X) = \sum_{x \in D} xp(x)$$

- The variance of X , denoted by $V(X)$ or σ_X^2 , is

$$V(X) = \sigma_X^2 = \sum_{x \in D} (x - \mu_X)^2 p(x) = E[(X - \mu_X)^2]$$

- Variance is a measure of spread
- Each distance from the mean gets weighted by the probability it will happen
- If we know all $p(x)$ then it is the true population variance for our RV, hence σ_X^2
- It can also be shown that $V(X) = E(X^2) - [E(X)]^2$
- Standard deviation is just the square root, σ_X

Variance and SD of Double Coin Toss



Double Coin Flip {HH, HT, TH, TT}

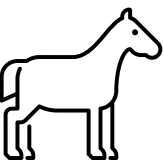
Let X = the number of tails

$$\left. \begin{array}{l} X(HH) = 0 \\ X(HT) = 1 \\ X(TH) = 1 \\ X(TT) = 2 \end{array} \right\} \begin{array}{l} P(X = 0) = 0.25 \\ P(X = 1) = 0.50 \\ P(X = 2) = 0.25 \end{array}$$

$$E(X) = 1$$

See slide 21

Variance of horse kicks



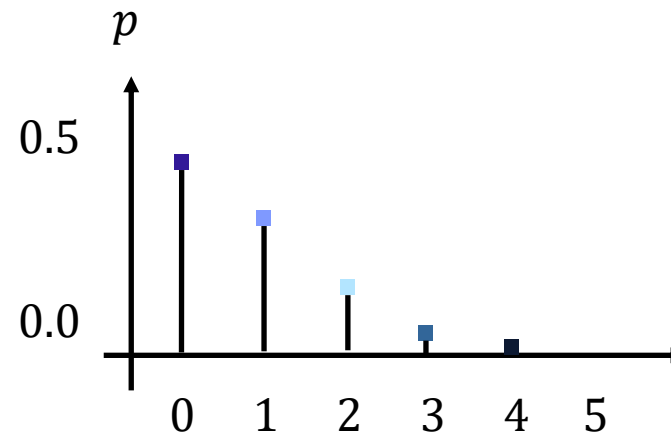
Total number of deaths by horse kick per year for a given Division

$\mu_X = 0.61$, See slide 21

$$\sigma_X^2 = \sum_{x \in D} (x - \mu_X)^2 p(x)$$

$$\begin{aligned} &= (-0.61)^2 \times 0.545 \\ &+ 0.39^2 \times 0.325 \\ &+ 1.39^2 \times 0.110 \\ &+ 2.39^2 \times 0.015 \\ &+ 3.39^2 \times 0.005 \end{aligned}$$

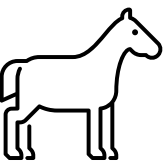
$$= 0.6079$$



$$\begin{aligned} \mu_X &= 0.61 \\ \sigma_X^2 &= 0.61 \end{aligned}$$

X	X-E(X)	N	p
0	-0.61	109	0.545
1	0.39	65	0.325
2	1.39	22	0.110
3	2.39	3	0.015
4	3.39	1	0.005

Variance of a Bernoulli RV



- Expected value of a Bernoulli Random Variable

$$\sigma_X^2 = \sum_{x \in D} (x - \mu_X)^2 p(x)$$

$$= (-p)^2 q + q^2 p$$

$$= pq(p + q)$$

$$= pq(p + 1 - p)$$

For Bernoulli RV
 $\sigma_X^2 = pq$

Bernoulli: does a soldier survive a kick?

Outcomes

A: Soldier is killed by a horse kick

B: Soldier is not killed by a horse kick

Bernoulli RV

$$X(A) = 1$$

$$X(B) = 0$$

$$\mu_X = p$$

Probabilities

$$P(A) = p$$

$$P(B) = 1 - p = q$$

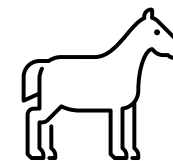
X	X-E(X)	P
0	-p	q
1	q	p



Binomial Distributions

A finite number of trials with binary outcomes

Revisiting Horse Kicks



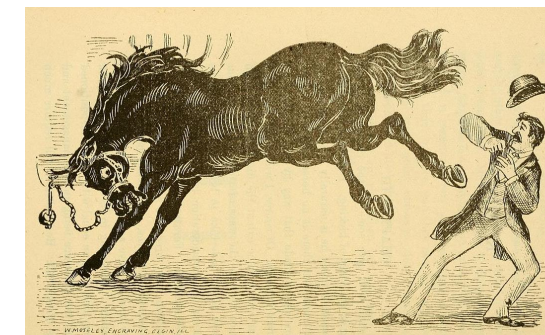
- N is the number of Division-Years
- X is number of deaths-by-horse
- Expected value is $\mu_X = 0.61$ (deaths/division- year)
- Variance is $\sigma_X^2 = 0.61$
- We need a model for the data



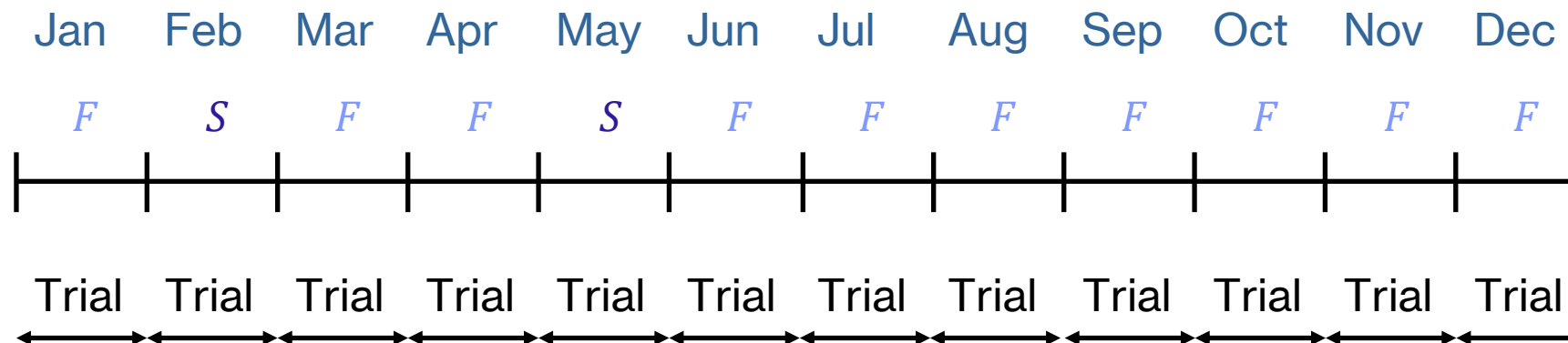
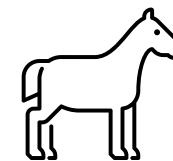
- Let us consider the model that the data are described by a Binomial Random Variable
- Consider each month of the year to be a “trial”

X	N	p
0	109	0.545
1	65	0.325
2	22	0.110
3	3	0.015
4	1	0.005

<https://www.biodiversitylibrary.org/pageimage/20321596>



Binomial Experiment



Success

S : at least 1 soldier dies by horse kick

$$P(S) = p$$

Failure

F : no soldier dies by horse kick

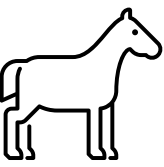
$$P(F) = 1 - p = q$$

n individual 'trials'



“A murderer that is a horse”

General Binomial Experiment



A **binomial experiment** is defined as follows:

1. The experiment consists of a sequence of n smaller experiments (trials)
2. Each trial can result in one of two possible outcomes, which we denote by S (success) and F (failure) – [Bernoulli RV]
3. The trials are independent from each other
4. The probability of success, p , is constant from trial to trial

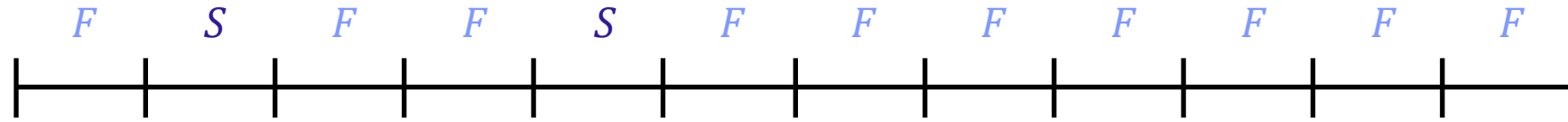
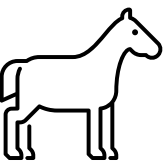
1. Each month

2. Soldier dies or no soldier dies

3. Have to assume no action is taken if a soldier does die

4. Have to assume that horse moods are at least repeatable

Binomial Random Variable



- X is the number of S across all trials
- In this case, $X = 2$, $n = 12$
- For a specific sequence, e.g.
 - $qpqqpqqqqqqq$
- The probability is
 - $p^x(1 - p)^{n-x}$
- Lots of ways to get to $X = 2$
- Order does not matter (combinations)

We can define the PMF of the Binomially distributed RV X as

$$b(x; n, p)$$

Where n and p are the parameters that describe the distribution

$$q = 1 - p$$

$$p^x q^{n-x}$$

To avoid confusion
vs the parameter p

Binomial Probability Distribution

- How many combinations are there?
- From n trials, we choose $X = x$
- For each possible outcome of X , x , we can use $C_{x,n}$ to give us the number of combinations
- $C_{x,n}$ is also known as the Binomial coefficient

x replaces k in this context

$$\binom{n}{x} = \frac{n!}{x! (n - x)!}$$

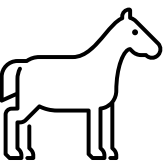
$$X \sim \text{Bin}(n, p)$$

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

X follows a binomial distribution with parameters n and p

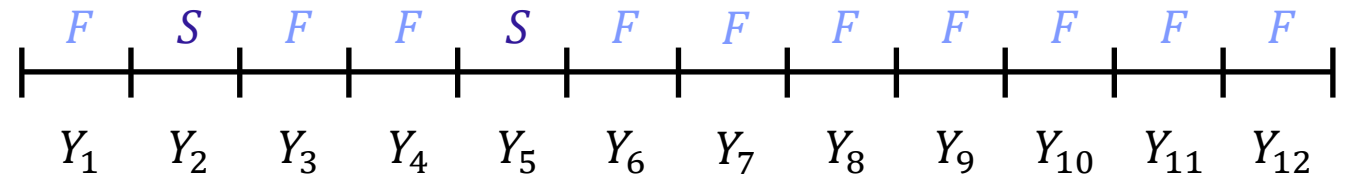
$$b(x; n, p) = P(X = x)$$

Expected Value of Binomial Distribution



- Binomial RV can be thought of as a sum of Bernoulli RV, Y_i
- Each Y_i represents 1 trial (number of deaths per month per division)
- X is the total number of deaths per year per division

$$X = \sum_{i=1}^n Y_i$$



$$\begin{aligned} \mu_X &= E\left(\sum_{i=1}^n Y_i\right) = E(Y_1 + Y_2 + \cdots + Y_n) \\ &= E(Y_1) + E(Y_2) + \cdots + E(Y_n) \end{aligned}$$

$$= \sum_{i=1}^n E(Y_i)$$

Recall
 $E(Y_i) = p$
for Bernoulli RV

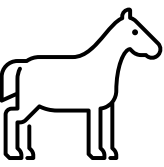
$$= \sum_{i=1}^n p = np$$

For binomial RV
 $\mu_X = np$

Light blue bar: derivation
not examinable

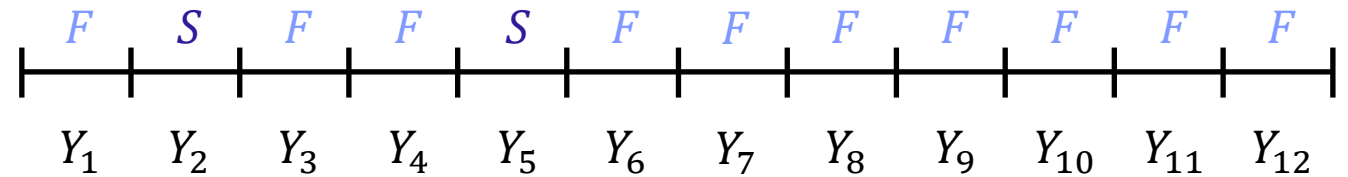


Variance of Binomial Distribution



- Binomial RV can be thought of as a sum of Bernoulli RV, Y_i
- Each Y_i represents 1 trial (number of deaths per month per division)
- X is the total number of deaths per year per division

$$X = \sum_{i=1}^n Y_i$$



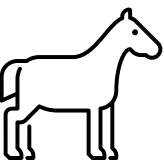
$$\begin{aligned}
 V(X) &= V\left(\sum_{i=1}^n Y_i\right) = V(Y_1 + Y_2 + \cdots + Y_n) \\
 &= V(Y_1) + V(Y_2) + \cdots + V(Y_n) \\
 &= \sum_{i=1}^n V(Y_i)
 \end{aligned}$$

Recall
 $V(Y_i) = pq$
 for Bernoulli RV

$$= \sum_{i=1}^n pq = npq$$

For binomial RV
 $\sigma_X^2 = npq$

Binomial Coefficient



$$X = 2, n = 12$$

$$\binom{n}{x} = \frac{n!}{(n-x)!x!}$$

MATLAB function for the binomial coefficient

Displays a message in the command window

```
%% Manual vs Auto generation of binomial coefficient
n=12;
X=2;
BinomCoeff_Man=factorial(n)/(factorial(n-X)*factorial(X));
BinomCoeff_Auto=nchoosek(n,X);
disp('Manual calculation yields '+ string(BinomCoeff_Man));
disp('Auto calculation yields '+ string(BinomCoeff_Auto));
```

Command Window

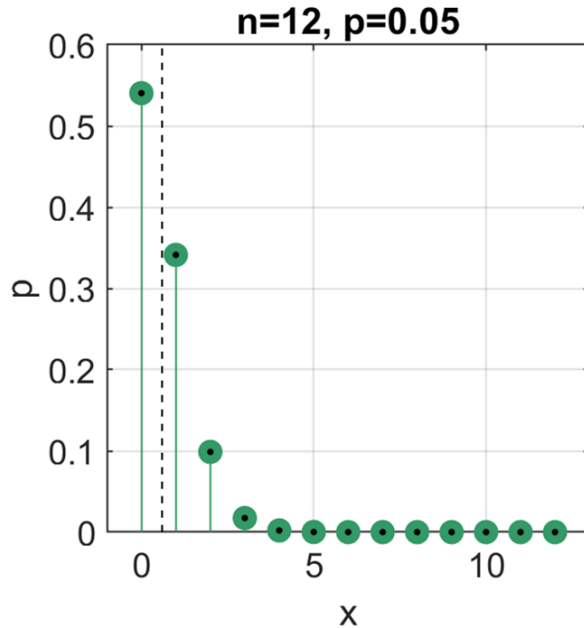
```
Manual calculation yields 66
Auto calculation yields 66
fx >>
```

New MATLAB function for appending strings

Can also use
['string 1', 'string 2']

Binomial Probability Distribution

$\cdot^*$ and $\cdot^{\wedge}$ do element-wise operations. See slide 48



```
%% Manual vs Auto generation of binomial pdf
p=0.05; % probability of success
% Generate the pdf
x=0:n; % Possible values of the Binomial RV
b_manual=factorial(n)./(factorial(n-x).*factorial(x)).*p.^x.*(1-p).^(n-x);
% Generate a binomial PDF manually
b_auto=binopdf(x,n,p); % Generate a binomial PDF using in-built MATLAB function
E=n*p; % Expected Value
```

MATLAB function for a binomial pmf

```
f1=figure(1);
set(f1,'position',[100 200 800 800],'color','w')
```

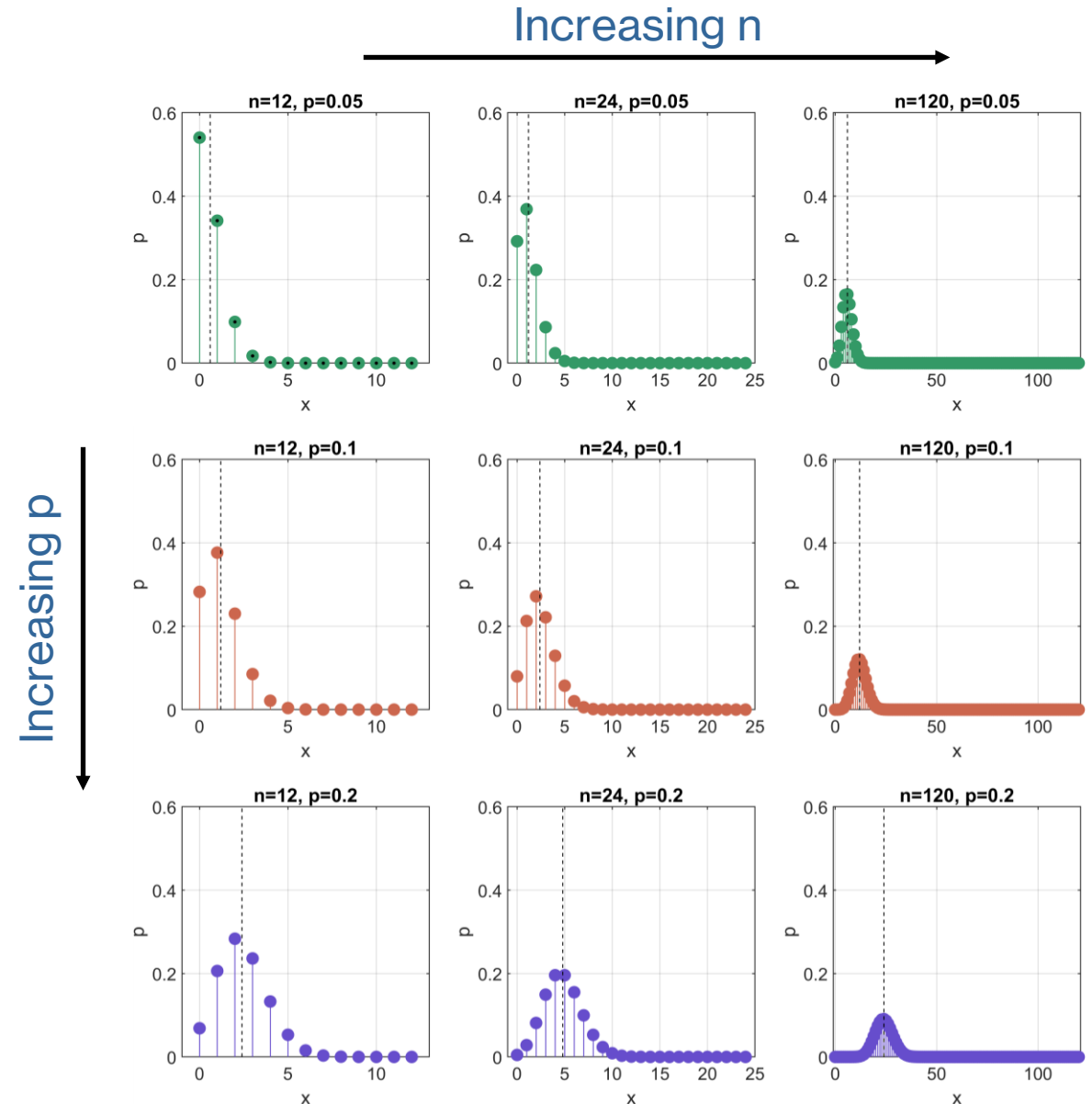
```
stem(x,b_manual,'filled','color',[0.2 0.6 0.4])
plot(x,b_auto,'k.')
plot(E*[1 1],[0 0.6],'k--')
xlabel('x');ylabel('p');axis([-1 n+1 0 0.6])
```

Plots a line between (x_1, y_1) and (x_2, y_2) using
`plot([x1 x2],[y1 y2])`

See Lecture_code for a grid of Binomial distributions with different n and p

Binomial Probability Mass Function

- See [Lecture_code](#)
- As n increases
 - More possible values of X , so the probability of a given value of x decreases
 - Note that sum of all values is 1 for all cases (note x -axis scale)
- As p increases,
 - Expected value of X increases
 - Variance of X increases
- Looks increasingly like a Normal distribution
- There is a Normal approximation to the Binomial distribution (and Poisson distribution) not much use as we have computers



Double Coin Toss as Binomial RV



Heads, F : $q = \frac{1}{2}$

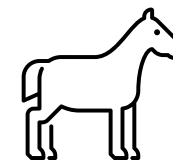
Tails, S : $p = \frac{1}{2}$

$$E(X) = np$$
$$V(X) = np(1 - p) = npq$$

Double Coin Flip {HH, HT, TH, TT}

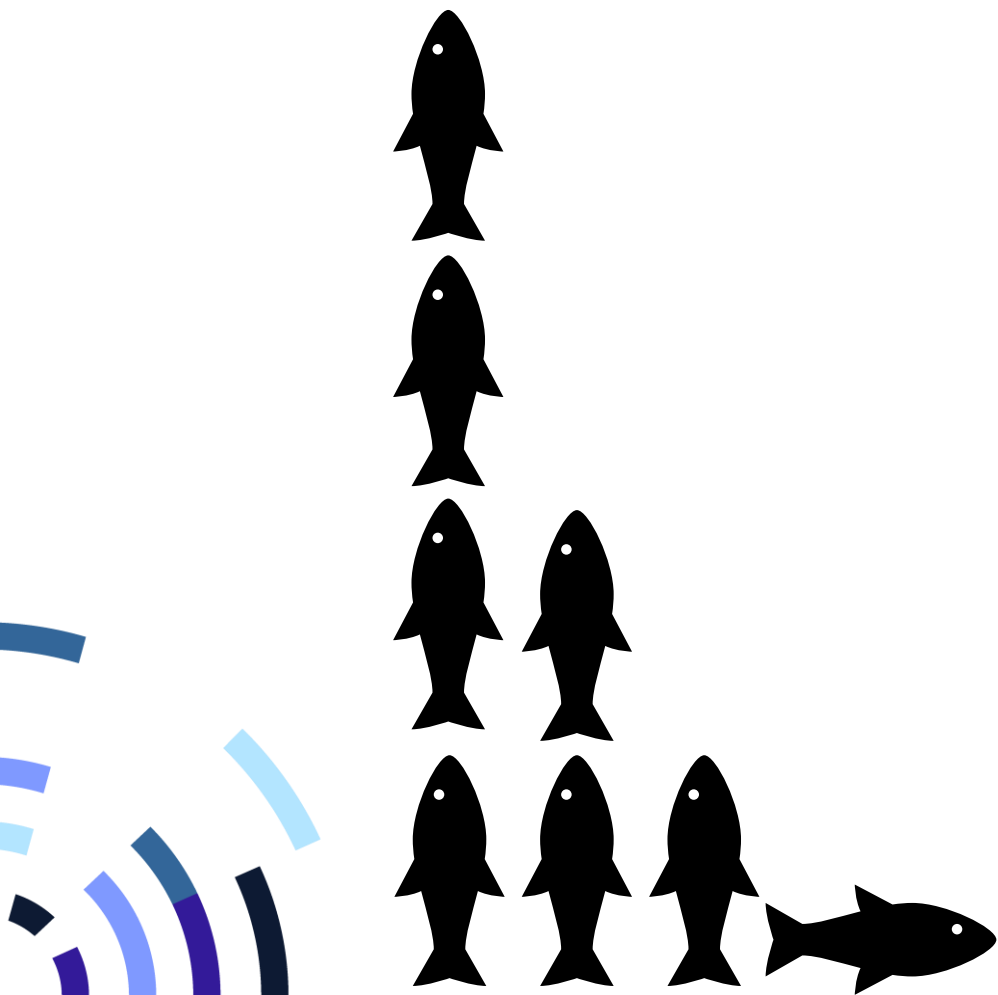
Note these are the
same values we
got on slides 21
and 26

Horse Kicks – but is a Binomial RV valid?



- We could have had two or more horse kicks in a given month!
- Hence point 2 in the definition of a binomial experiment is not met
- **“2. Each trial can result in one of two possible outcomes, which we denote by S (success) and F (failure)”**
- How short would the trial length have to be for Binomial RV to be appropriate?
- Weeks, Minutes, Seconds, Milliseconds?
- If we let the trial length tend to zero (e.g. a trial is a second) then point 2 can be achieved

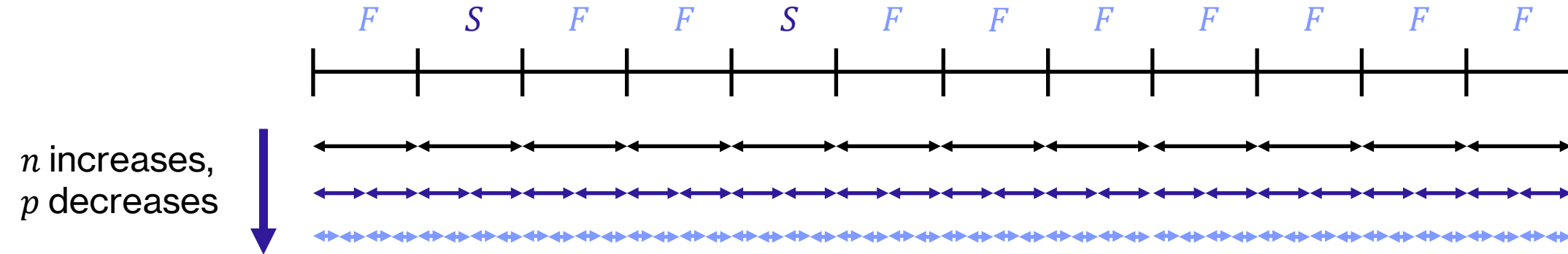
X	N	p
0	109	0.545
1	65	0.325
2	22	0.110
3	3	0.015
4	1	0.005



Poisson Distribution

A continuous number of trials with discrete outcomes

Poisson Probability Distribution



E.g.

- $n = 12, p = 0.1$
- $n = 24, p = 0.05$
- $n = 48, p = 0.025$
- Generally, $p \propto 1/n$

$$p = \frac{\lambda}{n}$$

- Where λ is a constant of proportionality

Binomial Distribution

$$\lim_{n \rightarrow \infty} P(X = x) = \lim_{n \rightarrow \infty} \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \left(\frac{\lambda}{n}\right)\right)^{n-x}$$

[Proof here if you are interested](#)

$$X \sim \text{Pois}(\lambda)$$

$$p(x; \lambda) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

Expected value and Variance of Poisson RV

Poisson can be thought of as a special case of Binomial where $n \rightarrow \infty$

$$p = \frac{\lambda}{n}$$

$$\mu_X = np = n \frac{\lambda}{n} = \lambda$$

Binomial

$$\mu_X = np$$

$$\sigma_X^2 = np(1 - p) = npq$$

$$\sigma_X^2 = n \frac{\lambda}{n} \left(1 - \frac{\lambda}{n} \right) = \lambda \left(1 - \frac{\lambda}{n} \right)$$

$$n \rightarrow \infty, \frac{\lambda}{n} \rightarrow 0$$

$$\sigma_X^2 = \lambda$$

Poisson

$$\mu_X = \lambda$$

$$\sigma_X^2 = \lambda$$

Poisson example vs Binomial

Binomial

- The outcome from a single trial can only be 0 or 1

Poisson

- One cannot define a trial for which the outcome from the trial can only be 0 or 1
- E.g.
 - there is on average 1 badger per square mile
 - what is the probability of finding at least 15 badgers in an area of 10 square miles?
 - $\lambda = 10$ badgers per 10 square miles
 - $p(x; 10) > 15 = 1 - (p(0; 10) + p(1; 10) \dots + p(14; 10)) = 0.05$
 - There is no area that could practically constitute a single trial





Binomial example vs Poisson

Binomial

- n finite independent trials, for which the outcome is 0 or 1
- X is total number of 'successes' or 1s
- e.g. a litter of 20 mice, from which we want homozygous alleles
 - Each mouse is a trial, success is homozygous
 - The probability of each mouse being homozygous is equal to 0.2
 - What is the probability of getting no homozygous offspring?

Binomial

Poisson Approximation

- Poisson approximation is not bad, but inaccurate (would get fewer marks)



Effect of choice of 'success'

- If instead we chose heterozygous as success, but still want probability of getting no homozygous offspring?

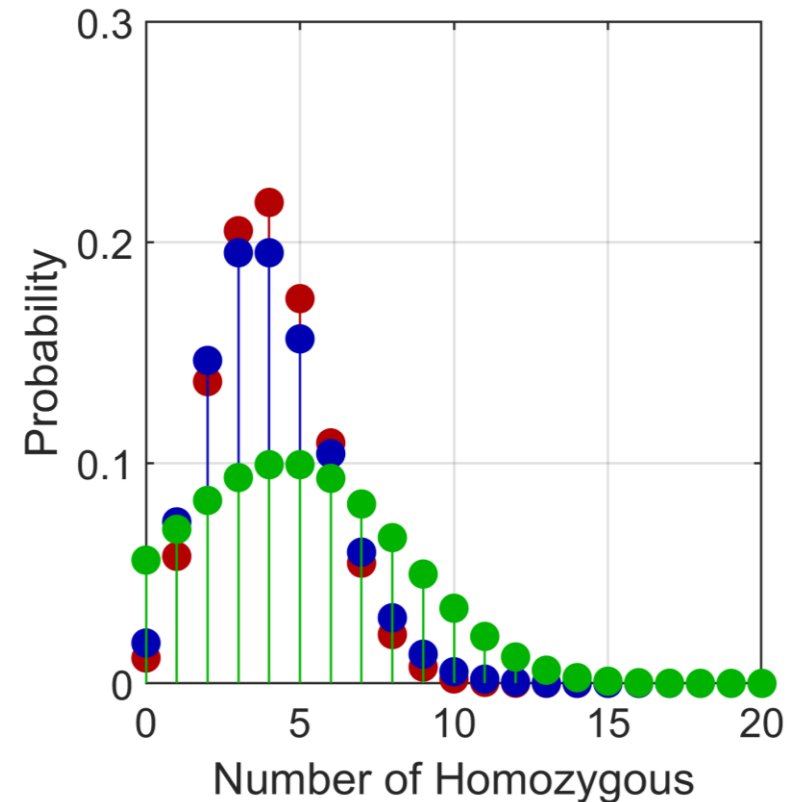
$$n = 20, x = 20, p = 0.8$$

$$b(20; 20, 0.8) = \binom{20}{20} 0.8^{20} (0.2)^0 = 0.012$$

$$\lambda = np = 0.8 \times 20 = 16$$

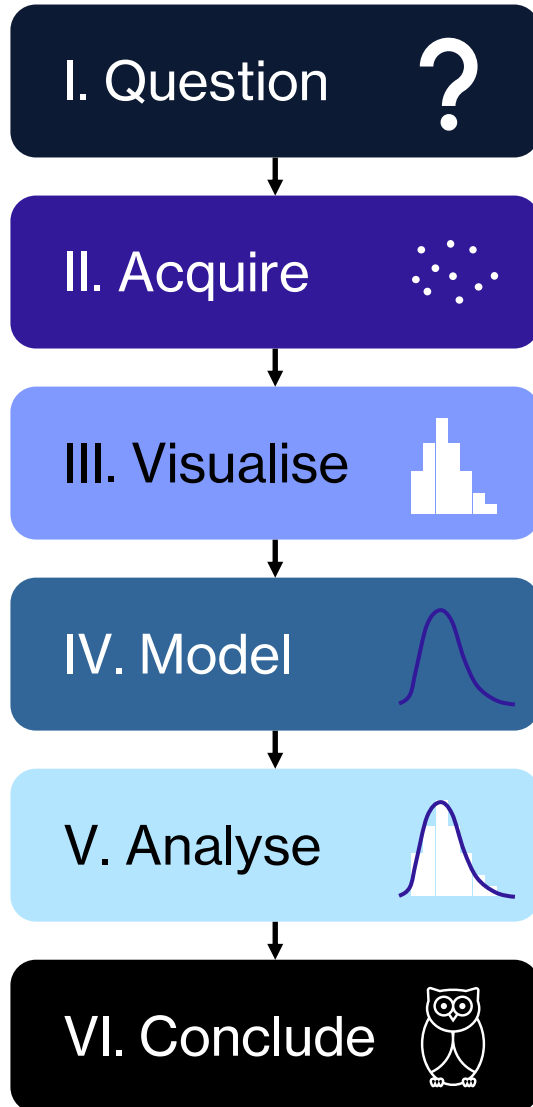
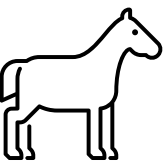
$$p(20; 16) = \frac{e^{-16} 16^{20}}{20!} = 0.056$$

- Binomial is the same, but Poisson approximation is much worse
- Because Poisson is special case of binomial with very small p
- Plot the PMFs against the number of homozygous (see Lecture_Code)



- Poisson model is worse for green model with heterozygous as success
- Generally, use the right model!

Framework – Vengeful Horses

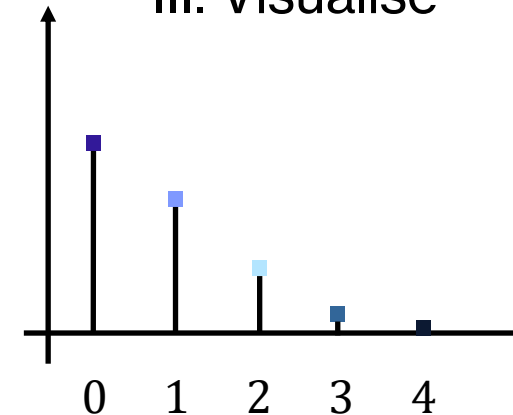


I. Is someone at fault for the deaths-by-horse?

II. Data

X	N
0	109
1	65
2	22
3	3
4	1

III. Visualise



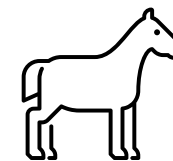
IV. The data are sampled from a Poisson distribution

– Check if the data are consistent with this model

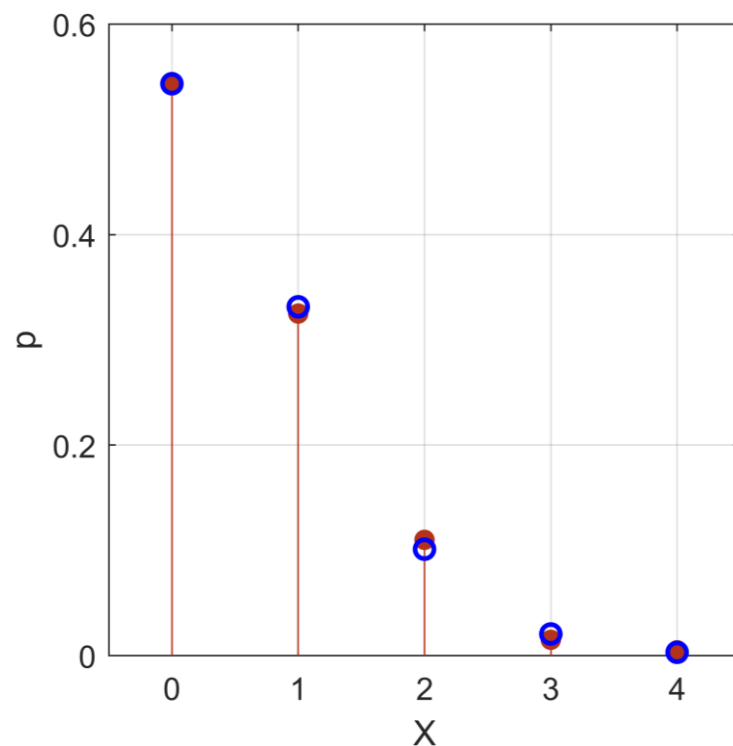
V. If they are, estimate how frequently we might expect 3 or 4 deaths-by-horse in a given division-year

VI. Compare these numbers to our data and draw a conclusion

Horse kicks as Poisson RV



- We need the value of λ , the parameter of the distribution: the 'average' number of deaths from horse kick over all division-years
- We know that this is $E(X)=0.61$ (from slides 22 and 44)



```
X=0:4;
```

```
N=[109 65 22 3 1];
```

```
p=N./sum(N);
```

```
E=sum(X.*p);
```

```
Lambda=E;
```

```
po_manual=exp(-Lambda).*Lambda.^X./factorial(X);
```

```
po_auto=poisspdf(X,Lambda);
```

```
stem(X,po_manual,'filled','color',[0.7 0.2 0.1])
```

```
plot(X,p,'bo','linewidth',1.5)
```

- Data are consistent with the model

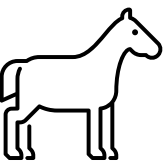
$\cdot$ * does element-wise multiplication
 $0.545 \times 0 + 0.325 \times 1 + 0.110 \times 2 + 0.015 \times 3 + 0.005 \times 4 = 0.61$

Calculate the model PMF

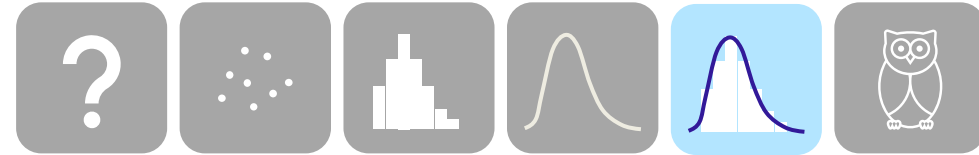
Plot the model

Plot the data

Probability of Horse kicks as Poisson RV



$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$



There are on average 0.61 deaths by horsekick for a given division-year
What is the probability of getting 3 or more deaths by horsekick in a given division $P(X \geq 3)$

$$P(X = 1) = \frac{e^{-0.61} 0.61^1}{1!} = 0.61e^{-0.61} = 0.331$$

$$P(X = 2) = \frac{e^{-0.61} 0.61^2}{2!} = \frac{0.61^2}{2} e^{-0.61} = 0.101$$

```
>> 1-poisscdf(2,0.61)

ans =

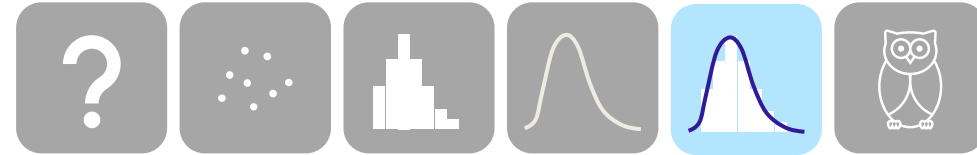
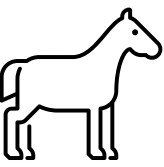
    0.0241
```

```
>> 1-(poisspdf(0,0.61)+poisspdf(1,0.61)+poisspdf(2,0.61))

ans =

    0.0241
```

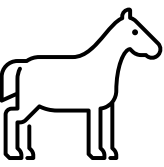
Number of Horse kicks as Poisson RV



Correspondingly, how many times might you expect this to happen in 200 division-years?

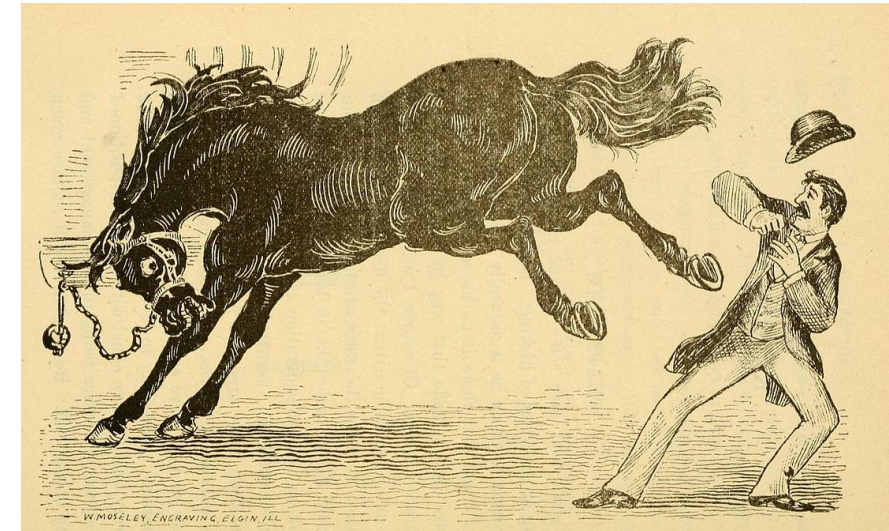
We observed 4 such cases

What have we learned?

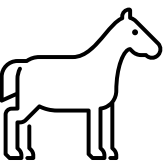


- Model check: the data is consistent with $X \sim \text{Pois}(\lambda)$ (our model is appropriate)
- Conclude: the case of 4 deaths in one year is not unexpected
- This implies that there is no-one specific fault
 - This does not mean that there is no fault!
 - Always take a good dose of common sense with your stats
- Rare events can happen
 - This will be important to keep in mind when we look at hypothesis testing later in the course

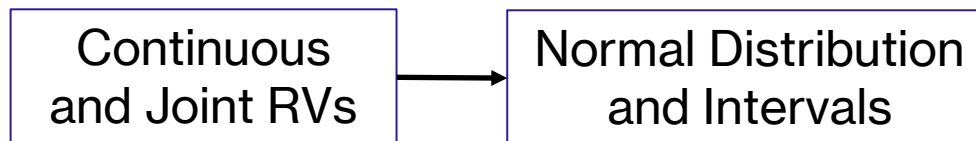
X	N (from data)	N (from model)
0	109	108.6
1	65	66.3
2	22	20.2
3	3	4.1
4	1	0.6



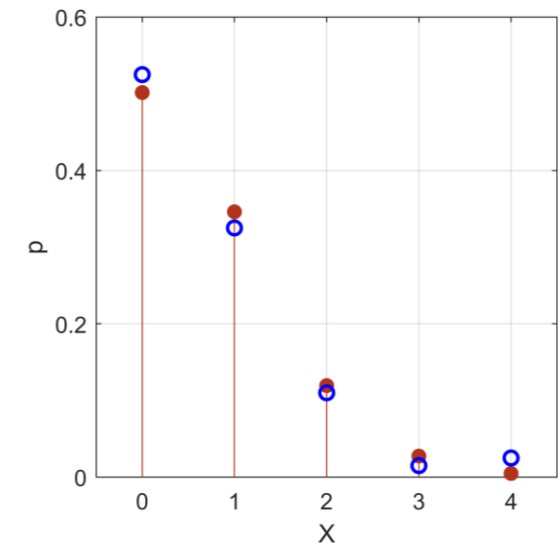
What would have implied fault?



- Consider a minor change in the difference
- $\lambda = 0.69$
- Model still looks reasonable overall, although definitely worse
- Model predicts 6.4 vs 8 cases with $X \geq 3$
- Intuition tells us something is wrong (lots of $X = 4$)
- We need formal approaches for evaluating the data and model
- This is where NHST comes in (Lecture 6), but first



X	N (from data)	N (from model)
0	105	100.3
1	65	69.2
2	22	23.9
3	3	5.5
4	5	0.9



3: Discrete Random Variables

What we covered

- Introduction to RVs
- Probability Mass Function
- Cumulative Distribution Functions
- RV Descriptive Statistics
 - Expected Value
 - Variance
- Binomial Distribution
- Poisson Distribution

Next Up

4: Continuous and Joint Random Variables

